

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

COMPUTER GRAPHICS

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SCOPE

FILLED AREA PRIMITIVES

Seed fill Overview

Boundary fill/edge fill algorithm

4 connected and 8 connected pixels approaches

Boundary fill/edge fill algorithm steps



SEED FILL ALGORITHMS

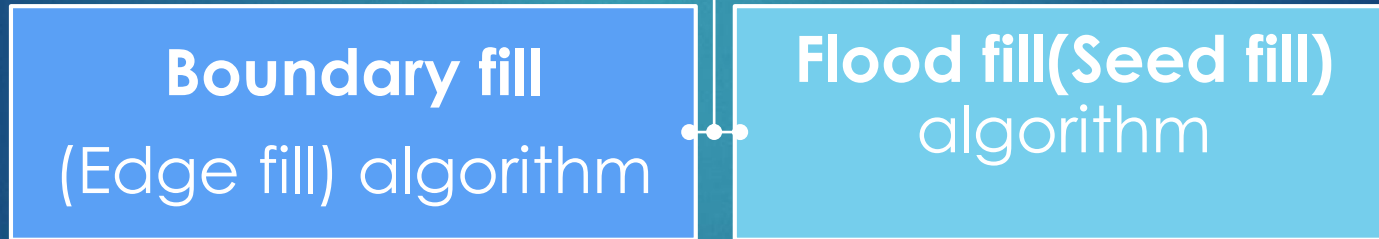


BOUNDARY FILL

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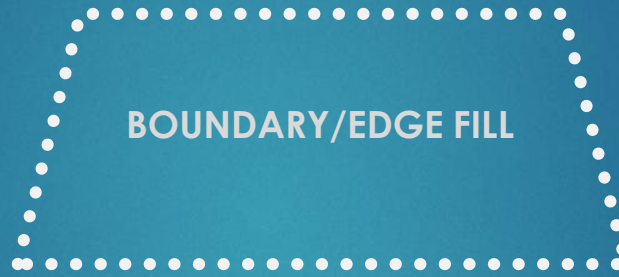
SEED FILLING OVERVIEW?

Is one of polygon filling approaches by starting from a given “seed” inside the polygon and highlight outward from this point to boundary pixels.



BOUNDARY FILLING ALGORITHM?

This algorithm picks a point inside an object and start to fill until it hits the boundary of the object.



BASICS

The color of the boundary and the color the filling area should be different for this algorithm to work.

Assume the color of the boundary is same for the entire object.

Starting with some seed, any point inside the polygon we examine the neighboring pixels to check whether the boundary pixel is reached

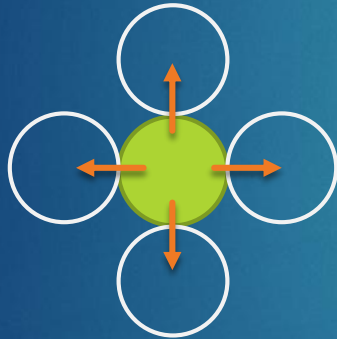
If the boundary pixels are not reached, pixels are highlighted and the process is continued until boundary pixels are reached.

The boundary fill algorithm can be implemented by 4-connected pixels or 8 connected pixels



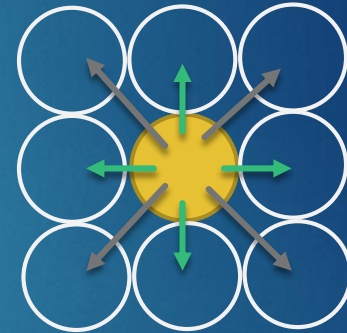
4 CONNECTED PIXELS OR 8 CONNECTED PIXELS APPROACHES

4 connected pixels or 8 connected pixels approaches are used to fill the boundary.



4 connected region

Every pixel can be reached
by a combination of moves
in FOUR directions:
(Left, Right, Up, Down)



8 connected region

Every pixel can be reached
by a combination of moves
in EIGHT directions:
(2 horizontals, 2 verticals
and 4 diagonals)



BOUNDARY FILL ALGORITHM STEPS

First initialize the 4 values:

x, y, Fill-color & Default-color.

Define the value of the boundary pixel color or Boundary color

Check if the current pixel is of Default-color and if **Yes** then Repeat **Step 4** and **step 5** till the boundary pixels are reached.

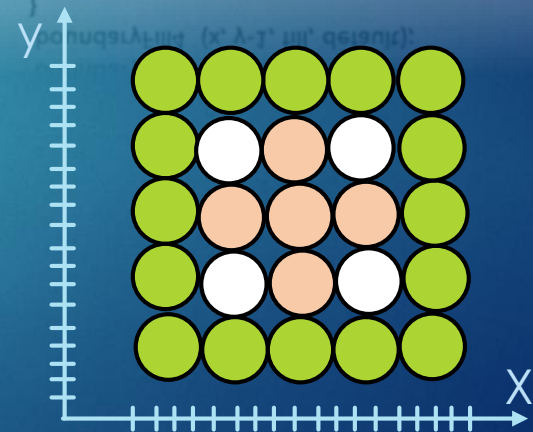
Repeat step 3 and step 4

for the neighboring 4 pixels.

Change the default color with fill color at the current pixel

```
void boundaryFill4 (int x, int y, int fill, int boundary, int default)
{
    int current;
    current = getpixel (x, y);
    if ((current != boundary) && (current = default))
    {
        setPixel (x, y, fill);
        boundaryFill4 (x+1, y, fill, default);
        boundaryFill4 (x-1, y, fill, default);
        boundaryFill4 (x, y+1, fill, default);
        boundaryFill4 (x, y-1, fill, default);
    }
}
```

4 Connected Fill Approach



BOUNDARY FILL ALGORITHM STEPS

First initialize the 4 values:

x, y, Fill-color & Default-color.

Define the value of the boundary pixel color or Boundary color

Check if the current pixel is of Default-color and if **Yes** then Repeat **Step 4** and **step 5** till the boundary pixels are reached.

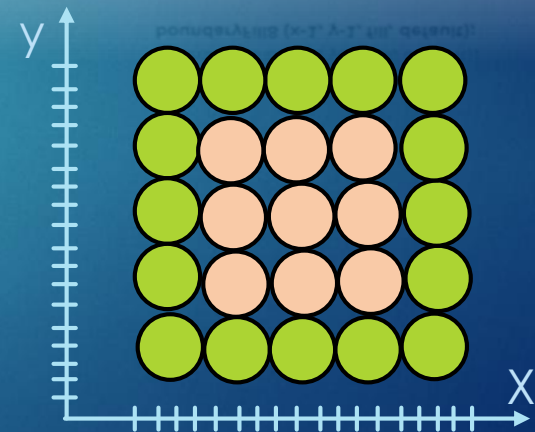
Repeat step 3 and step 4

for the neighboring 4 pixels.

Change the default color with fill color at the current pixel

```
void boundaryFill8 (int x, int y, int fill, int boundary, int default)
{
    int current;
    current = getpixel (x, y);
    if ((current != boundary) && (current == default))
    {
        setPixel (x, y, fill);
        boundaryFill8 (x+1, y, fill, default);
        boundaryFill8 (x-1, y, fill, default);
        boundaryFill8 (x, y+1, fill, default);
        boundaryFill8 (x, y-1, fill, default);
        boundaryFill8 (x+1, y+1, fill, default);
        boundaryFill8 (x+1, y-1, fill, default);
        boundaryFill8 (x-1, y+1, fill, default);
        boundaryFill8 (x-1, y-1, fill, default);
    }
}
```

8 Connected Fill Approach





**MORE PRACTICE MAKES
YOU PERFECT**





Always Seek Knowledge



THE END

